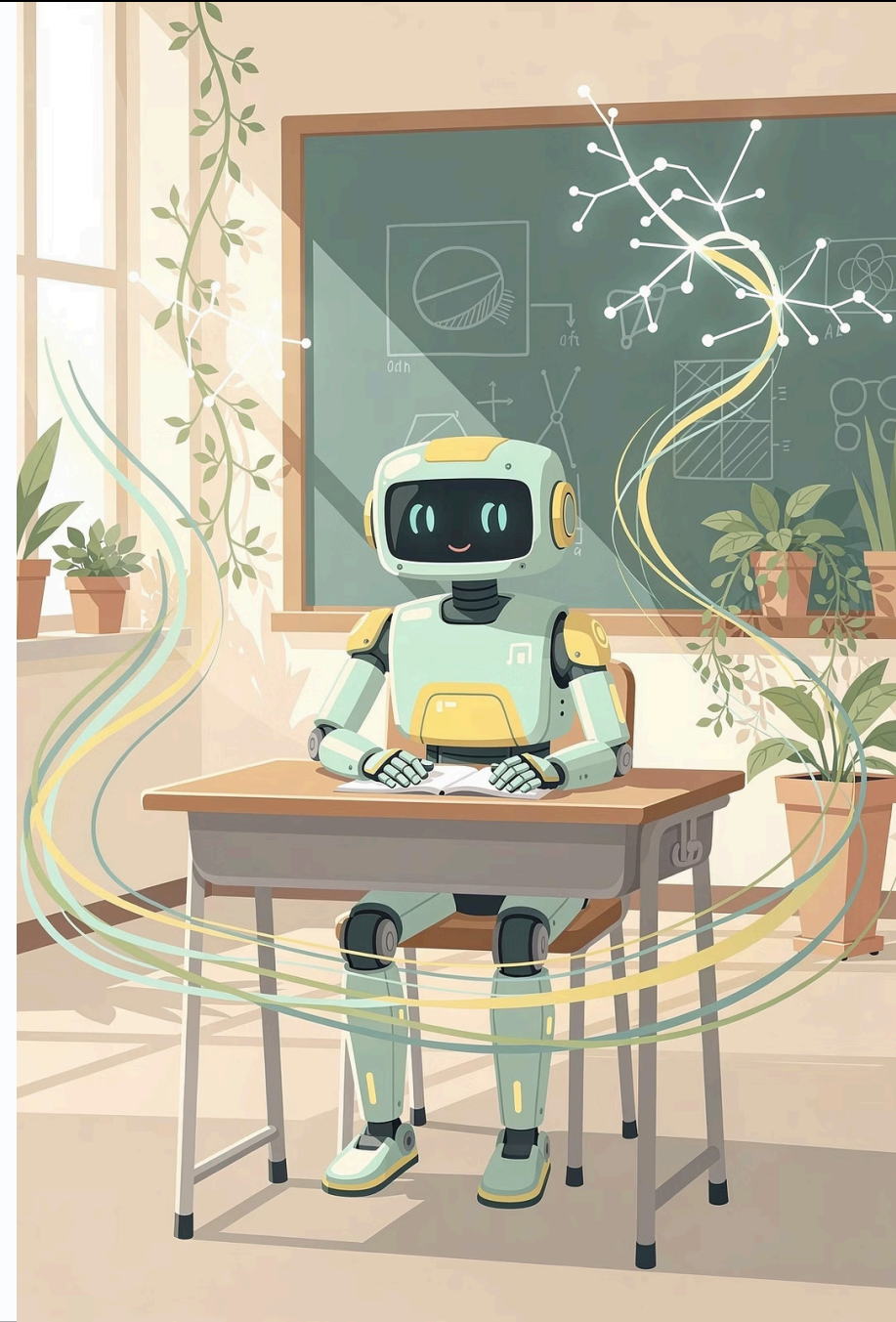


Session 3:

What is Machine Learning?

Today we will discover how computers actually learn from examples. You've heard the term "Machine Learning", now you'll finally understand what it really means and how it's different from traditional programs.

Get ready!



Recap : What We Learned in Session 2

Last time, we learned the building blocks of AI. Let's refresh our memory:

Concept	What It Means
Data	Information AI learns from – photos, text, numbers, sounds
Algorithm	Step-by-step instructions AI follows – like a recipe
Training	The process where AI learns from data – like studying flashcards
Model	The result of training – a skilled "expert" that can make predictions
Prediction	AI's smart guess on new data – based on patterns it learned

📋 **The AI Recipe** : Data + Algorithm → Training → Model → Prediction

Today we dive deeper into the second step of that recipe: **Machine Learning - the actual "learning from data" part!**

Objectives

By the end of this session, you will be able to:



Define Machine Learning

Explain what ML means in your own words



Compare Approaches

Tell the difference between traditional programs and machine learning



Understand Why Rules Fail

Explain why rule-based systems can't handle everything



Spot Patterns

Recognize how ML finds patterns in data



Identify ML in Your Life

Find examples of ML in apps you use daily



See ML in Action

Watch a real ML model learn in real time with Google Teachable Machine



Big Question of the Day

How do computers actually learn from examples?

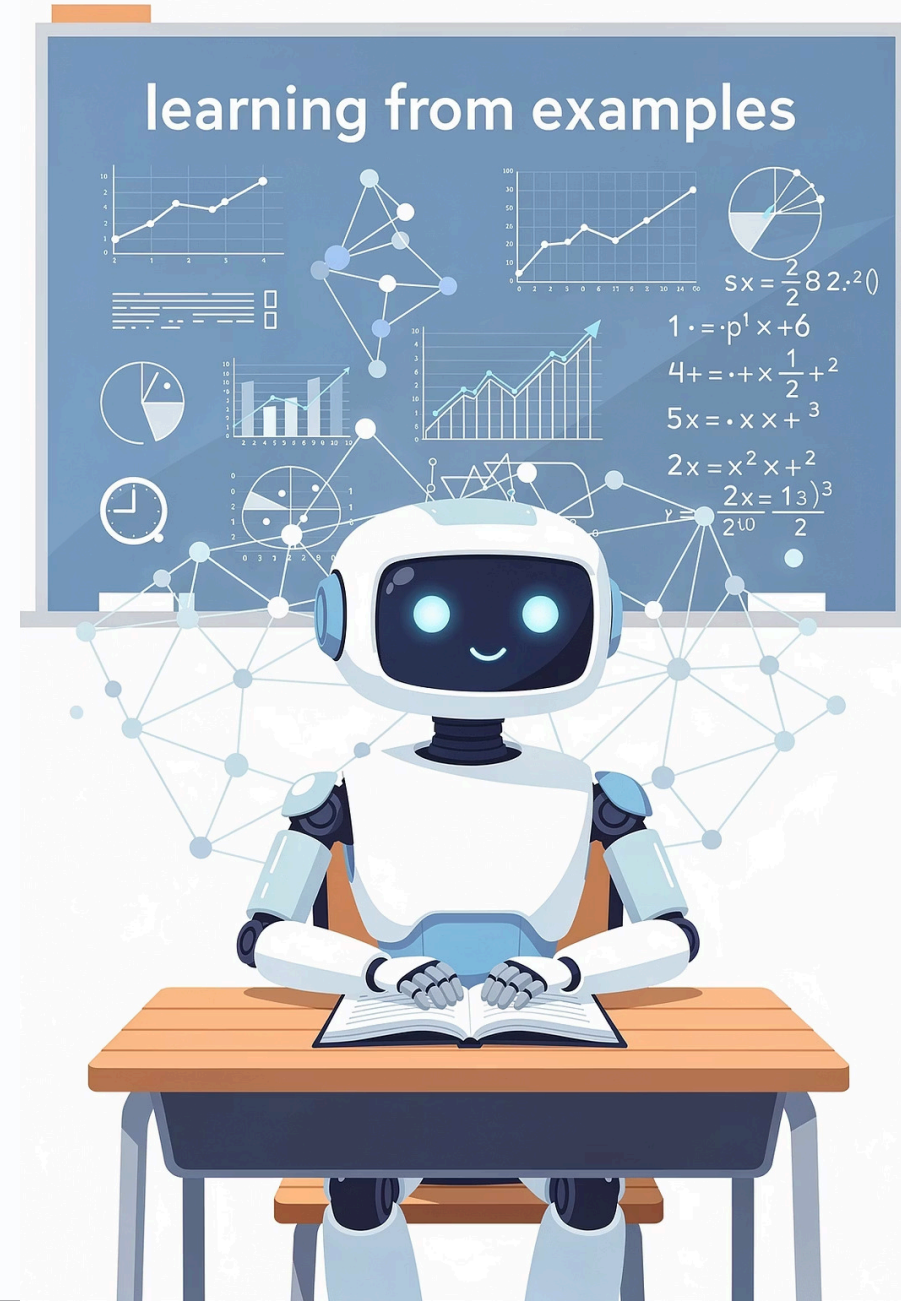
Think about it: When you learned to recognize a dog, someone probably showed you a dog few times and said "dog." But computers need thousands of examples. Why? How does it work?

By the end of this session, you'll know the answer!

What is Machine Learning?

Machine Learning (ML) is a way for computers to learn patterns from data instead of following fixed rules written by humans.

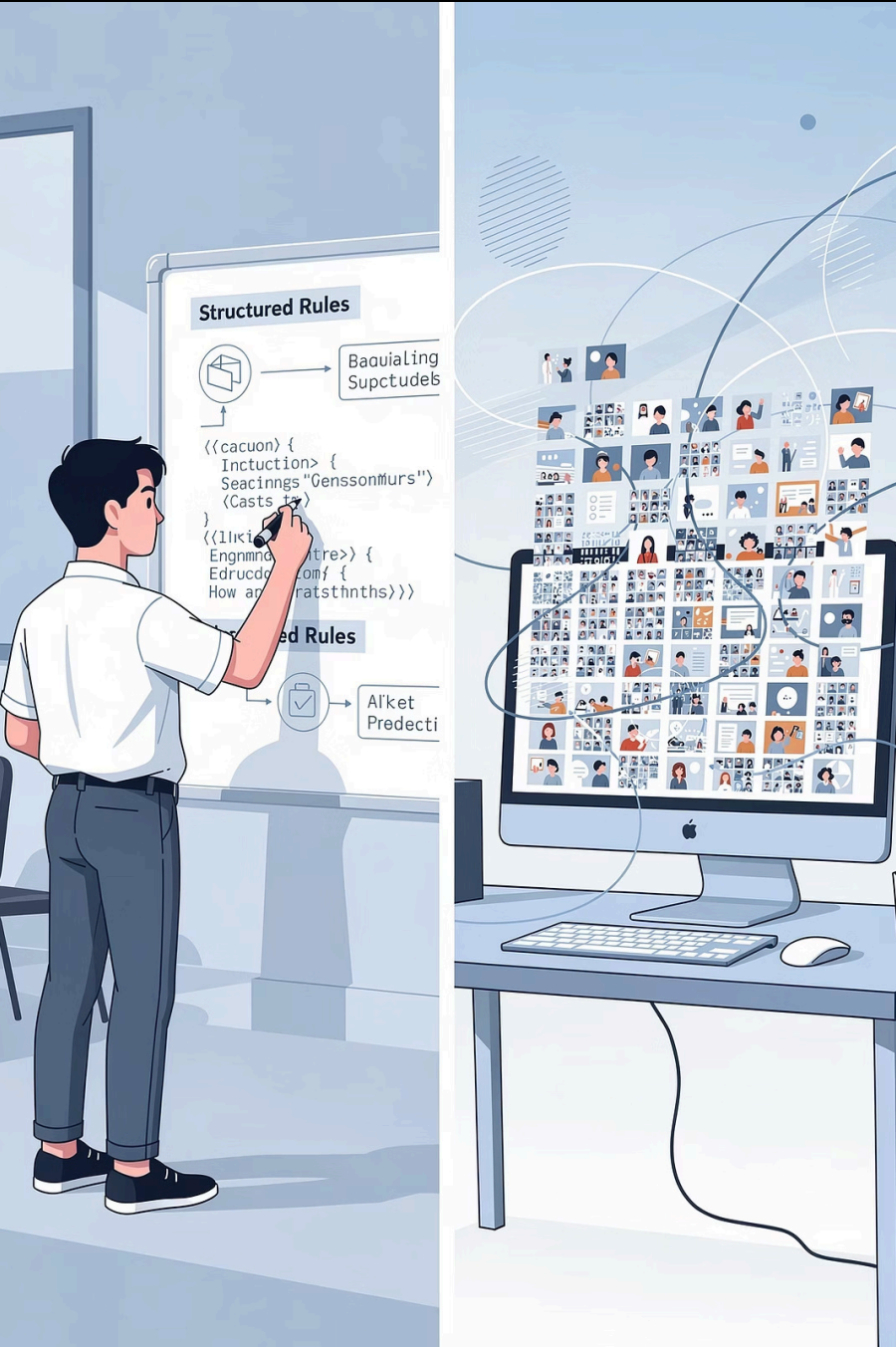
In technical terms : Machine Learning is a subset of artificial intelligence where computers use data to learn and improve on their own, without being explicitly programmed for every single situation.



Think of It This Way:

Traditional Programming	Machine Learning
You give the computer rules	You give the computer examples
Computer follows instructions	Computer finds patterns
Computer does exactly what you told it	Computer learns what to do on its own
Works for simple, predictable tasks	Works for complex, changing situations

 **The key insight:** Instead of telling the computer what to do, we show it how to figure things out for itself!



Traditional Programming vs. Machine Learning

Traditional Programming

- You give the computer rules
- Computer follows instructions
- Computer does exactly what you told it
- Works for simple, predictable tasks

Machine Learning

- You give the computer examples
- Computer finds patterns
- Computer learns what to do on its own
- Works for complex, changing situations

Traditional Programs : Following Rules

Most software you use every day is traditional programming. This means a human wrote step-by-step instructions for every possible situation.

Examples You Know:

Program	What It Does	How It Works
Calculator	Adds, subtracts, multiplies, divides	Follows math rules written by programmers
Traffic Light	Changes from green to yellow to red	Timer-based rules: "30 seconds green, then 3 seconds yellow, then 30 seconds red"
Microwave	Cooks food for set time	Button press → timer counts down → beep
Alarm Clock	Rings at set time	"If time = 7:00 AM, then ring"

These programs are **reliable and predictable**. A calculator never decides to add differently today than it did yesterday. They just follow instructions.

Why Rules Can Fail

Here's the problem: **You can't write rules for everything!**

The Spam Filter Challenge

Imagine you're a programmer asked to write rules for a spam filter. You might start with:

Rule 1: If email contains "FREE MONEY!!!" → mark as spam
Rule 2: If email contains "You've won a prize" → mark as spam
Rule 3: If email is from unknown sender → mark as spam

But then spammers adapt:

- They write "F R E E M O N E Y" with spaces
- They use "Congratulations! You're a winner!"
- They pretend to be someone you know

Your rules keep failing because spammers invent new tricks. You'd have to write millions of rules and update them constantly - **an impossible task!**



Why We Need Machine Learning

Task	How Many Rules Would You Need?
Detect spam emails	Millions – spammers invent new tricks daily
Recognize a cat in a photo	Infinite – cats come in every pose, lighting, breed
Understand sarcasm in text	Impossible – depends on tone, context, relationship
Recommend a movie you'll like	No fixed rules – everyone has different taste

This is why we need Machine Learning. **Instead of writing rules for everything, we let the computer learn the patterns from examples.**

Mini Activity : Rule-Based or Learning-Based?

Let's practice telling the difference! For each system below, decide: Does it follow fixed rules written by humans? (**Rule-Based**) OR does it learn patterns from data? (**Learning-Based**)

System	Your Guess	Why?
Calculator		
YouTube recommendations		
Traffic light		
Spam filter in Gmail		
Microwave with preset buttons		
Face unlock on your phone		
Google Maps traffic prediction		
Alarm clock app		
Netflix suggesting a show		
Thermostat set to 72°		

Discuss: Which ones were hardest to decide? Why?

Mini Activity : Answer Key

System	Type	Why?
Calculator	Rule-Based	Follows fixed math rules written by programmers
YouTube recommendations	Learning-Based	Learns from what you watch and what others watch
Traffic light	Rule-Based	Runs on timers or simple sensors – no learning
Spam filter in Gmail	Learning-Based	Learns from millions of emails marked spam/not spam
Microwave with preset buttons	Rule-Based	Just follows programmed cooking times
Face unlock on your phone	Learning-Based	Trained on thousands of photos of YOUR face
Google Maps traffic prediction	Learning-Based	Learns from historical traffic data + real-time GPS
Alarm clock app	Rule-Based	Simple rule: "If time = set time, ring"
Netflix suggesting a show	Learning-Based	Learns from what you and similar users watched
Thermostat set to 72°	Rule-Based	Simple rule: "If temp < 72, heat on; if temp > 72, heat off"

 **Key Insight:** If it can learn and improve over time, it's likely Learning-Based. If it does the exact same thing every time, it's probably Rule-Based.

Pattern Recognition - How ML Finds Patterns in Data

At its core, Machine Learning is about one thing : **finding patterns in data**.

What Are Patterns?

Patterns are regular, repeating relationships in information. Your brain finds patterns constantly:

- You notice that every time you wear your lucky socks, your team wins (pattern!)
- You realize that when your teacher says "this might be on the test," it usually is (pattern!)
- You see that dark clouds often mean rain soon (pattern!)

Machine Learning does the same thing - but with massive amounts of data and patterns too complex for humans to spot.

Patterns in Your Data



Your Watch History

"People who watched this also watched that"



Emails Marked as Spam

"Spam emails often have these words, sender patterns, and formatting"



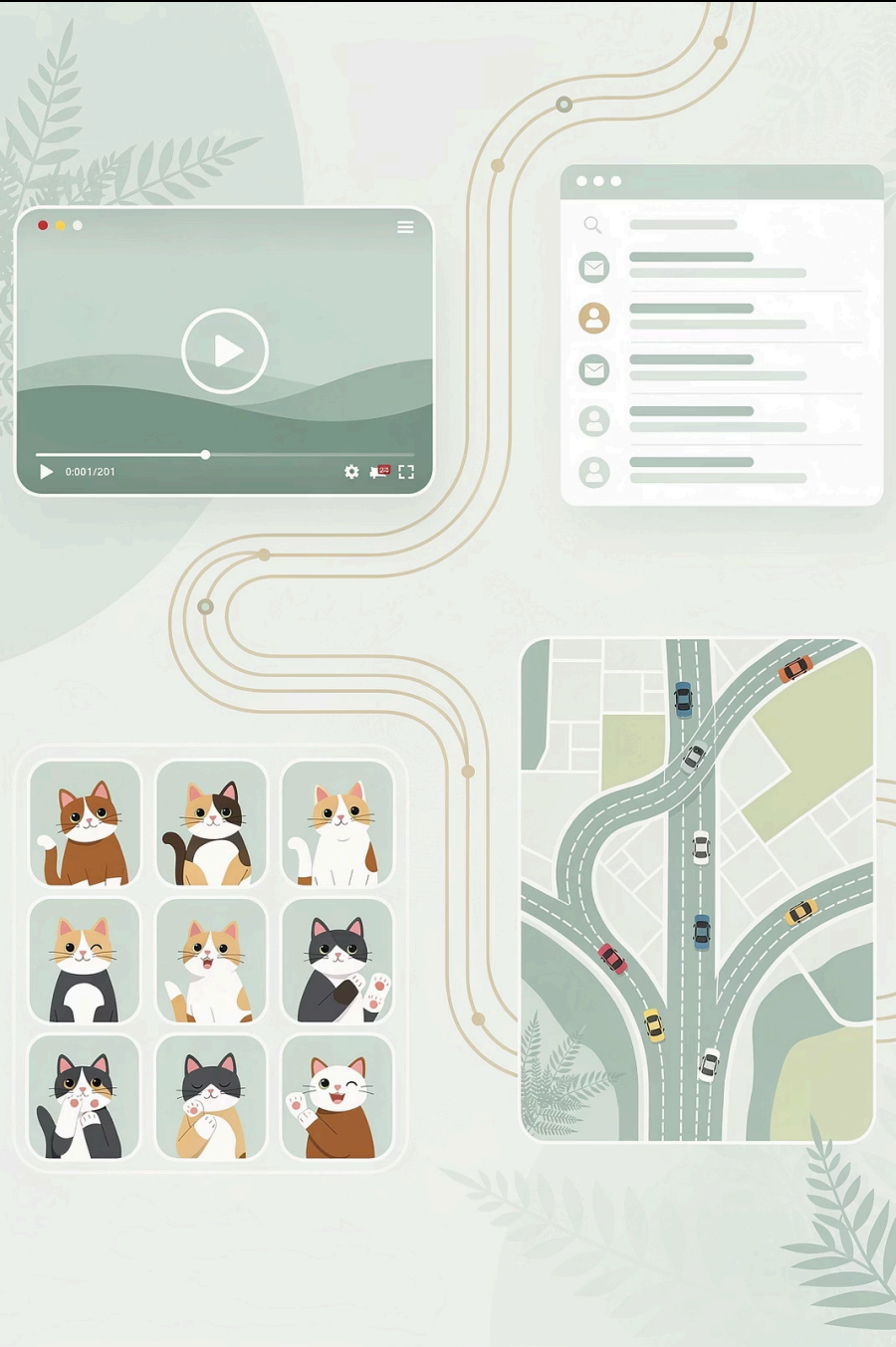
Photos of Cats

"Cats have pointy ears, whiskers, and certain eye shapes in specific arrangements"



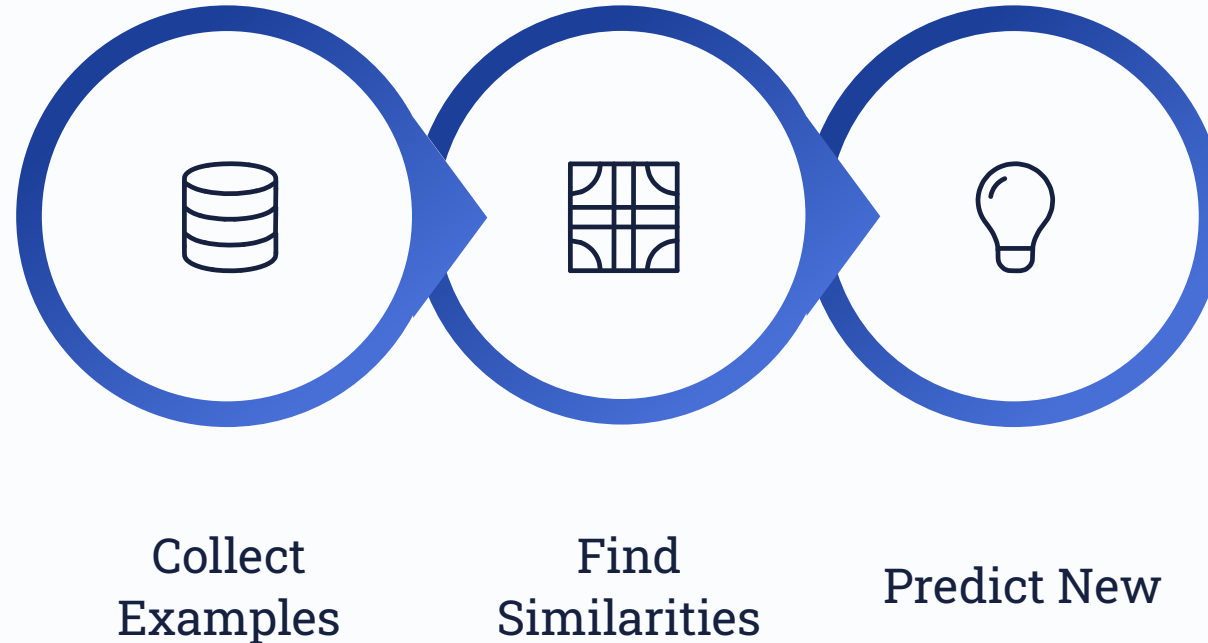
Traffic Data

"On Fridays at 5pm, this highway is usually congested"



How Pattern Recognition Works

Think of it like this:



This simple three-step process is the foundation of pattern recognition in machine learning.

A Simple Example: Happy Faces

Imagine you want ML to learn what a "happy face" emoji looks like:

Training Examples	What Patterns ML Finds
😊	Eyes: normal, Mouth: curved UP
😊	Eyes: normal, Mouth: curved UP
😄	Eyes: squinty, Mouth: wide UP
😁	Eyes: squinty, Mouth: wide UP with teeth

Pattern learned: Happy faces have mouths curved UPWARD!

Now when shown a new face 😊, ML predicts: "Mouth curved UP? Yes! **That's a happy face!**"

Learning from Patterns

Once ML finds patterns, it **learns** them. This means it can now recognize those patterns in **NEW data** it's never seen before.

The Training Process

1 Training Data (with labels)

Thousands of examples with correct answers: "This is a dog" "This is NOT a dog"

2 Algorithm (finds patterns)

The learning method - looks for patterns: "What's similar in all the dog photos?"

3 Model (the result)

The trained pattern-recognizer: "Now I know what dogs look like!"

4 New Photo (never seen)

A photo the AI has NEVER seen before: "Is this a dog or not?"

5 Prediction (smart guess)

Based on learned patterns, AI predicts: "90% confident - this IS a dog!"

Mini Activity 2 : Spot the Pattern!

Let's practice finding patterns - just like ML does!

Part A: Number Patterns

What comes next in each sequence? How do you know?

Sequence	Next Number?	What's the Pattern?
2, 4, 6, 8, __		
5, 10, 15, 20, __		
1, 4, 9, 16, __		
3, 6, 12, 24, __		

Part B: Shape Patterns

Draw or describe what comes next:

Pattern	Next Shape?
○, △, ○, △, —	
□, □, ○, □, □, ○, —	
●, ●, ●, ●, ●, —	

Part C: Trickier Patterns

Sequence	What comes next?
Sunny, Sunny, Rainy, Sunny, Sunny, Rainy, __	
Dog, Cat, Dog, Cat, Dog, __	
A, B, D, G, K, __	

Reflect: How did you figure these out? What "clues" were you looking for? That's EXACTLY what ML does – but with much more complex patterns!

Mini Activity 2 : Answer Key



Part A: Number Patterns

- 2, 4, 6, 8, **10** (add 2 each time)
- 5, 10, 15, 20, **25** (add 5 each time)
- 1, 4, 9, 16, **25** (square numbers: 1^2 , 2^2 , 3^2 , 4^2 , 5^2)
- 3, 6, 12, 24, **48** (multiply by 2 each time)

Part B: Shape Patterns

- ○, △, ○, △, ○ (alternating)
- □, □, ○, □, □, ○, □ (two squares, then circle, repeat)
- ●, ●, ●, ●, ●, ● (alternating colors)

Part C: Trickier Patterns

- Sunny, Sunny, Rainy... **Sunny** (two sunny, one rainy, repeat)
- Dog, Cat, Dog, Cat, Dog, **Cat** (alternating)
- A, B, D, G, K, **P** (skip increasing letters: +1, +2, +3, +4, +5)

📌 **Key Insight:** You just did what Machine Learning does! You found patterns in data and used them to predict what comes next. ML does this with billions of data points and patterns too complex for humans to spot.



Machine Learning in Everyday Life

You use ML all the time without even realizing it! Here are examples you already know:

Social Media

TikTok

Learns what videos you watch fully, what you skip, what you rewatch - then predicts what you'll want to see next

Instagram

Learns which posts you like, comment on, and spend time looking at - then arranges your feed

Snapchat Filters

Learns to detect your face, eyes, mouth, and movements - then tracks filters as you move

🎵 Music & Video

Spotify

Learns what songs you listen to, skip, and add to playlists – then creates Discover Weekly playlists just for you

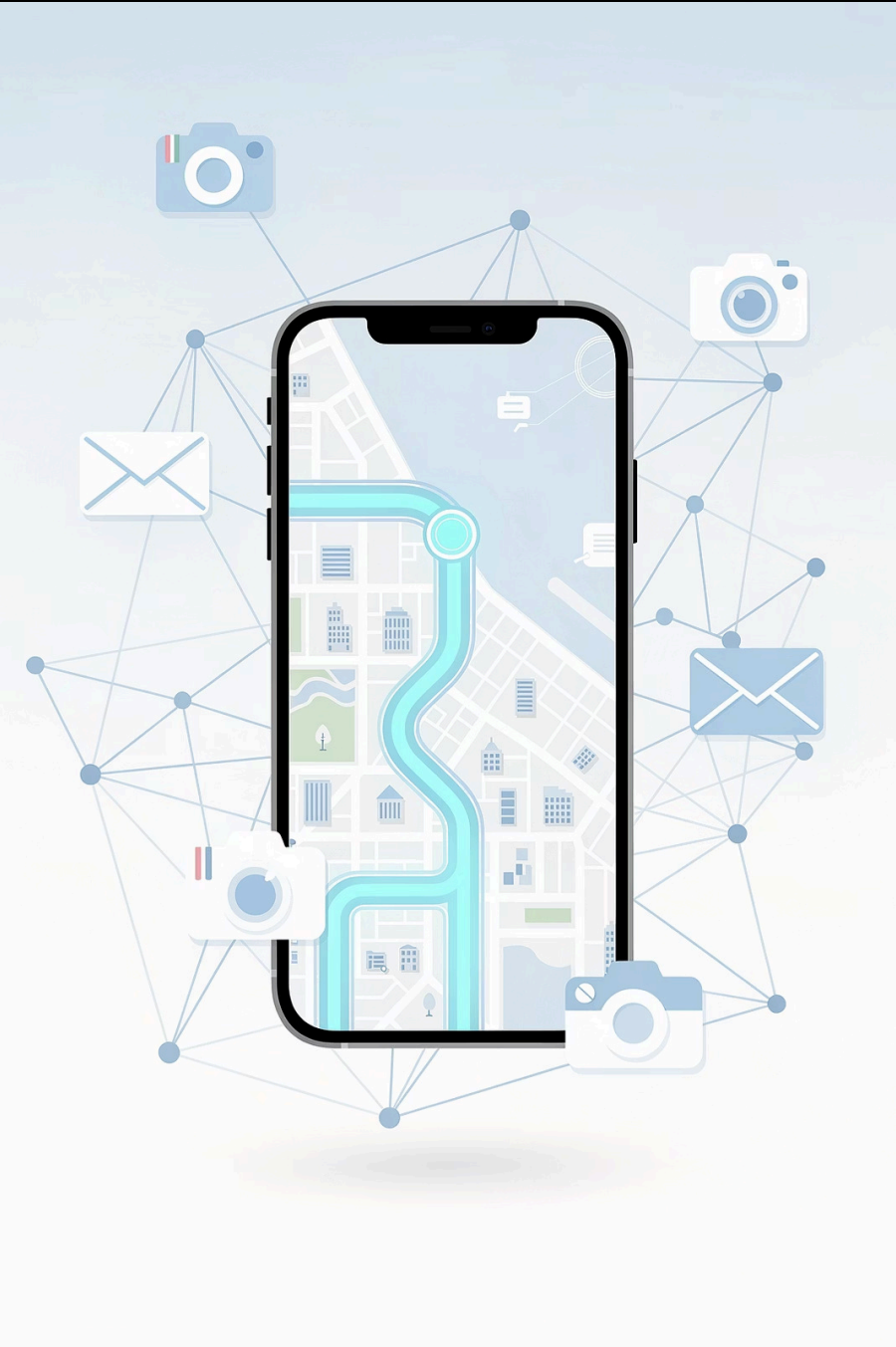
Netflix

Learns what you watch, when you stop watching, what you rate highly – then recommends shows you might like

YouTube

Learns from your watch history, searches, and what similar users watch – then suggests videos





Maps & Navigation

Google Maps

Learns from millions of phones' GPS data and historical traffic patterns then predicts your travel time and suggests fastest routes

Waze

Learns from real-time user reports about accidents, police, and traffic then reroutes drivers

Communication

Gmail Spam Filter

Learns from billions of emails marked spam/not spam then predicts which new emails are unwanted

Google Translate

Learns from millions of translated documents then predicts translations for new text

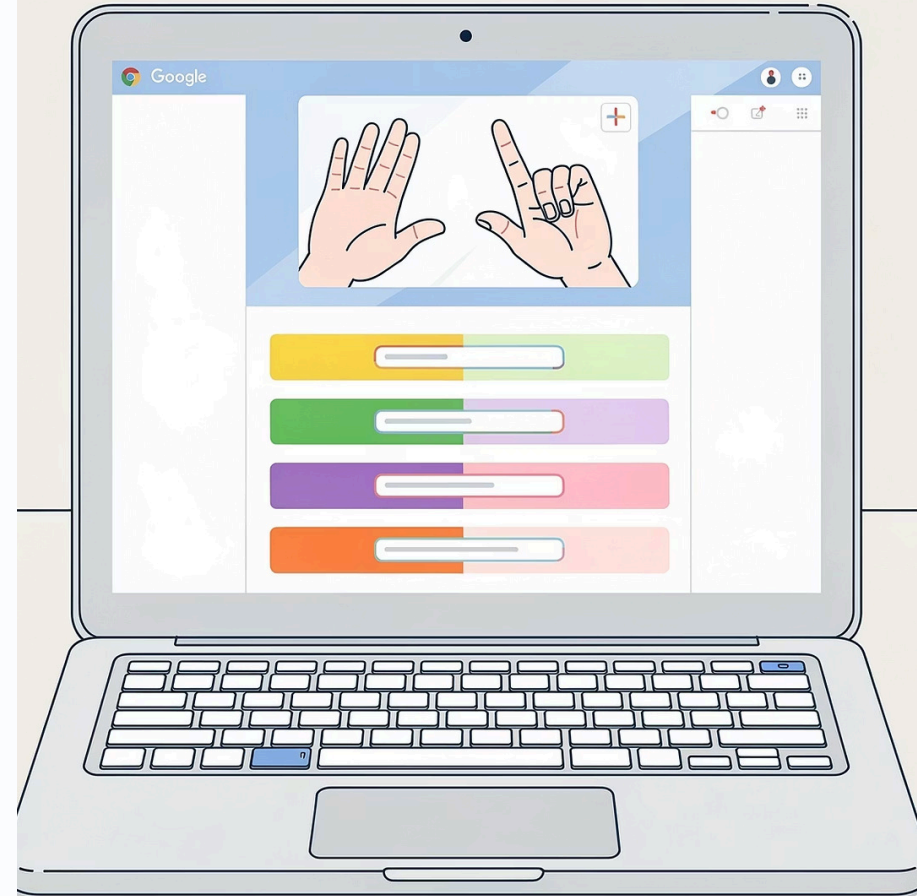
Hands-On Activity: Introduction to Google Teachable Machine

Now it's time to see Machine Learning in action! Today we'll just watch and learn - no building yet. Think of this as a "demo day" where you get to see what ML looks like behind the scenes.

What is Google Teachable Machine?

Teachable Machine is a free, easy-to-use tool that lets you train your own machine learning models **without writing any code**. It shows you exactly how ML works - you give it examples, it learns patterns, and then it can recognize new examples.

We'll use this tool throughout the course, starting in **Session 6** where you'll build your OWN model. Today, we'll just explore what it can do.



What We'll See Today

We're going to train a simple model that recognizes hand gestures:

Class 1: 🕊️ Peace Sign

The model will learn what a peace sign looks like

Class 2: 🖐️ High Five

The model will learn what a high five hand looks like

Class 3: 👊 Fist

The model will learn what a fist looks like

Then we'll test if the model can recognize NEW examples of these gestures - even if they're slightly different!

Step 1: Opening Google Teachable Machine

Watch as I open **teachablemachine.withgoogle.com**

PEACE

HIGH FIVE

FIST



Step 2: Collecting Training Examples

Now I'll add examples for each class. Watch what happens:

Class 1 (Peace Sign) 🍷

1

- Hold up a peace sign to the camera
- Click "Hold to Record" and capture 10-15 images
- Change angles: tilted, closer, farther
- Change lighting: brighter, darker, different backgrounds

2

Class 2 (High Five) 🖐

Same process – 10-15 examples from different angles

3

Class 3 (Fist) 👊

Same process – 10-15 examples from different angles



Key Point: The more VARIETY in examples, the better the model will learn. If we only show peace signs from one angle, it might not recognize peace signs from other angles!

Step 3: Training the Model

Now for the magic! click "**Train Model**" and we'll watch what happens

What's happening inside the computer:

- The algorithm looks at every pixel in every image
- It finds patterns: "Peace signs have two fingers up, high fives have five fingers spread, fists have curled fingers"
- It adjusts thousands of internal "knobs" (parameters) to recognize these patterns
- It repeats this process dozens of times, getting better each pass

This takes just a few seconds! Real ML models can train for **weeks or months**.

Step 4: Testing the Model

The model is trained! Now let's test it with NEW examples it's never seen.

Watch the Preview Window:

Gesture to Show	What the Model Predicts	Confidence
🕊️ Peace sign (new angle)	"Peace Sign"	98%
👋 High five (different lighting)	"High Five"	95%
👊 Fist (with funny background)	"Fist"	97%
👉 "Other" gesture	????	???

📘 **Confidence Score:** A percentage showing how sure the AI is about its prediction.

Remember: High confidence doesn't mean it's correct, the AI can be very confident and still be wrong!

What happened with the "Other" gesture?

The model hasn't seen this before! It might:

- Predict the closest match (maybe Peace Sign because it has two fingers)
- Show lower confidence (maybe 60% Peace, 30% Fist, 10% High Five)
- Get confused!

📄 This shows that models can **ONLY recognize patterns they were trained on.**

Let's Try to Confuse the Model!

Now I'll try gestures that might trick it:

Test	What Happens?	Why?
Peace sign but hand is far away	Still recognizes	Learned that hand size varies
Peace sign with different background	Still recognizes	Learned that background doesn't matter
Peace sign but only two fingers visible	Still recognizes	Learned the key pattern
"Rock On" gesture (👉)	Might confuse with Peace Sign	Similar finger pattern
Hand with fingers partially hidden	Lower confidence	Missing some of the pattern

Discussion: What might make the model fail completely?

What We Learned from the Demo

Observation	What It Teaches Us
The model needed 10-15 examples per class	ML needs MULTIPLE examples to learn
Variety in examples helped	Good data = good model
Training took seconds	But real models train for weeks/months
It recognized new examples	That's the goal – generalization!
It got confused by unfamiliar gestures	ML only knows what it was trained on
Confidence scores showed certainty	ML can be unsure – and that's okay!

📌 **Most Important Takeaway:** The model didn't "understand" peace signs or high fives. It doesn't know what they mean or why people make them. It just found patterns in pixels that correlate with the labels we gave it. This is the difference between **pattern recognition** and **true understanding**.

Summary



Machine Learning

A way for computers to learn patterns from examples instead of following fixed rules



Rule-Based System

A program that follows step-by-step instructions written by humans (like a recipe)



Pattern Recognition

Finding repeating patterns in data – the main way machine learning works

Summary: What We Learned Today

Concept	Key Idea
What is Machine Learning?	Computers learning patterns from examples instead of following fixed rules
Traditional Programming	Following step-by-step instructions written by humans (calculators, traffic lights)
Why Rules Fail	Can't write rules for everything - spam changes, cats look different, taste varies
Pattern Recognition	ML finds repeating patterns in data - just like you do in number sequences
Learning from Patterns	Once patterns are found, ML can recognize them in NEW data it's never seen
ML in Everyday Life	TikTok, Spotify, Netflix, Gmail, Google Maps - all use ML constantly
Teachable Machine Demo	We watched a model learn to recognize hand gestures from examples

The Most Important Idea

Machine Learning = Finding Patterns in Examples

Everything else - all the complex math, all the fancy tools, all the impressive results - comes back to this simple idea.

When you use :

- Instagram, ML finds patterns in what you watch.
- Spotify, ML finds patterns in what you listen to.
- Google Maps, ML finds patterns in traffic data.
- Face Unlock, ML finds patterns in photos of YOUR face.

Patterns. Examples. Learning. That's Machine Learning.





Homework: ML Detective



Your ML Hunt

- Find 3 systems you use daily that use Machine Learning.
- For each one, note: the system/app, what it does, and what patterns it learned.

Ideas: YouTube, Spotify, Netflix, Gmail, Google Maps, Snapchat, Instagram, Face Unlock, Amazon



Reflect

Which of your 3 examples would be hardest to build with rules instead of ML? Why?



Bonus Challenge (Optional)

Explain Machine Learning to someone at home — a parent, sibling, or friend. Can they explain it back to you!

Next Session Preview: Session 4 - Types of Machine Learning

Now that you know WHAT machine learning is, next time we'll discover the different ways machines can learn!



Supervised Learning

Learning with answer keys



Unsupervised Learning

Finding patterns alone



Reinforcement Learning

Learning by trial and error